Zoho Schools for Graduate Studies

Notes

Date : 25/09/2025

Day : Thursday

Multiple Inheritance

1. What is Multiple Inheritance?

Multiple Inheritance means a class can inherit features (fields and methods) from **more than one parent class**.

2. Does Java Support's Multiple Inheritance (with Classes)?

Ans: No, Java designers intentionally **disallowed multiple inheritance of classes** to avoid the **"Diamond Problem"**.

Reasons why java doesn't support multiple inheritance :

1. Deadly Diamond Problem (Ambiguity)

- If two parent classes have the **same method**, and a child class inherits both:
  - The compiler cannot decide which version to use.
  - This creates **ambiguity** and leads to confusion in method resolution.

2. Complexity in Design

- Handling multiple parent classes complicates the **compiler design** and the **object model**.
- Java was designed to be **simple and easy to understand** → avoiding multiple inheritance keeps the language clean.

3. Constructor Invocation Issues

- In multiple inheritance, both parent classes may have constructors that need to be called.
- Deciding the **order of constructor calls** becomes tricky and can lead to inconsistency.

Sample code for Multiple Inheritance

Game.java

```java
public class Game extends Object {
   Game() {
      System.out.println("Game");
   }
   void play() {
      System.out.println("Game started...");
   }
}
```

IndoorGame.java

```java
public class IndoorGame extends Game {
   IndoorGame() {
      System.out.println("Indoor Game");
   }
   void play() {
      System.out.println("Indoor Game started...");
   }
}
```

OutdoorGame.java

```java
public class OutdoorGame {
   void play() {
      System.out.println("Outdoor Game started...");
   }
}
```

Chess.java

```
// ❌ Trying to extend two classes (not allowed in Java)
public class Chess extends IndoorGame, OutdoorGame {
    Chess() {
        System.out.println("Chess");
    }
    void play() {
        System.out.println("Chess Game started...");
    }
}
```

InheritanceDemo.java

```
public class InheritanceDemo {
    public static void main(String[] args) {
        Chess c = new Chess();
        c.play();
    }
}
```

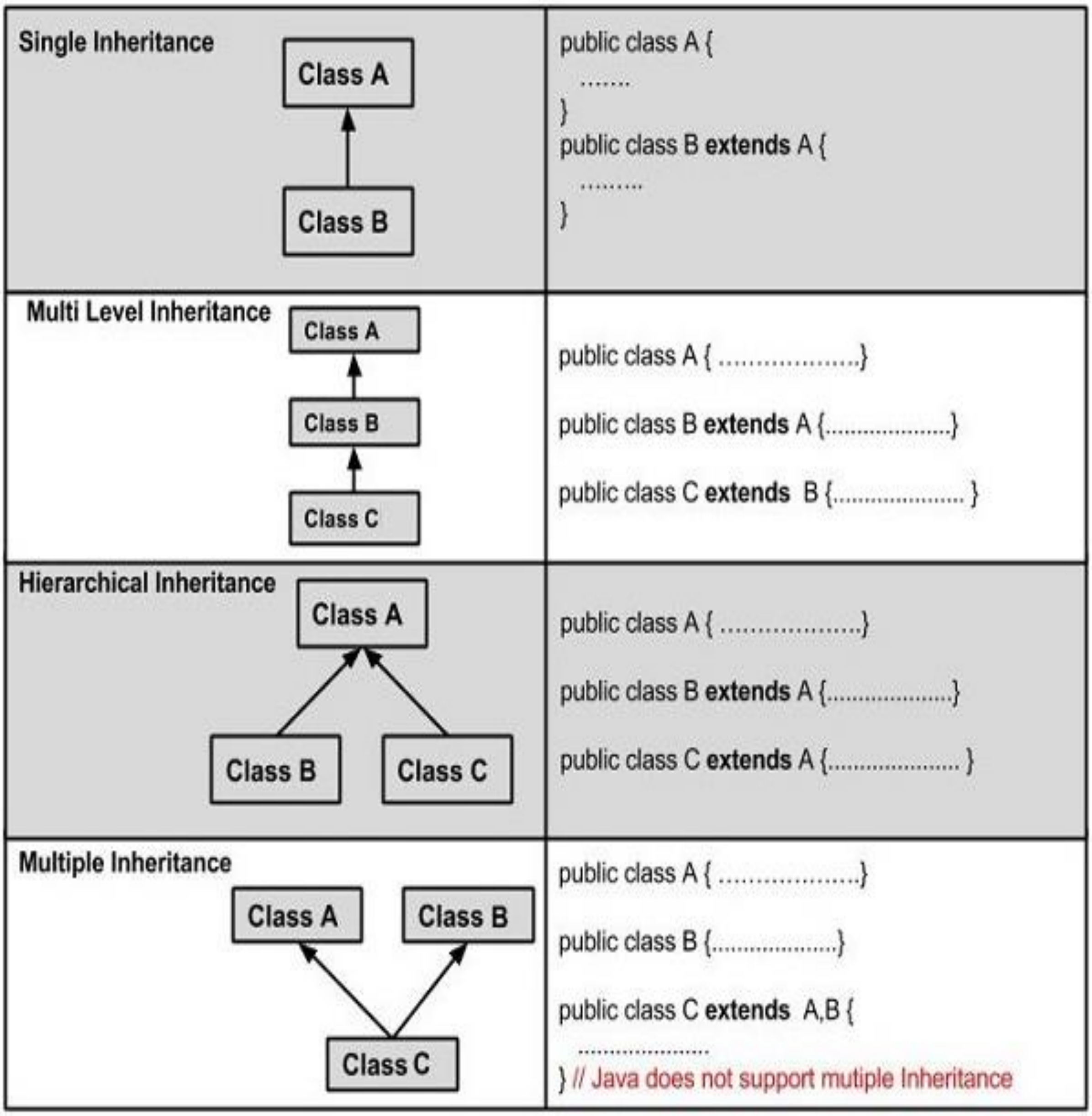
Output :

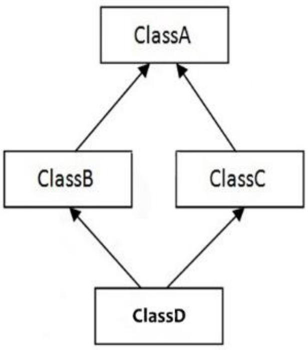
❌ Compile-time Error

When you try to compile `Chess.java`, you'll get:

```
error: '{' expected
public class Chess extends IndoorGame, OutdoorGame {
                                      ^
error: class, interface, or enum expected
public class Chess extends IndoorGame, OutdoorGame {
```

➢ Different Forms of Inheritance in Java





4) Hybrid Inheritance

➢ Rules for Representing Inheritance Diagrams

1. Use of Boxes for Classes

   - Each class should be drawn as a box/rectangle.
   - Write the class name clearly inside the box (sometimes with methods/attributes if needed).

2. Direction of Inheritance

   - Use a Strong straight line with an arrow to show inheritance.

- The arrow points from the child class (subclass) to the parent class (superclass).

Example :

Class A	**<----**	**Class B**
(Parent)		(Child)

1. Single Inheritance :

- One subclass should have only one arrow pointing upward to its parent.

2. Multilevel Inheritance :

- The subclass of one class can itself be a parent for another class.
- Represented as a vertical chain of arrows.

3. Hierarchical Inheritance :

- A single parent class can have multiple child classes.
- Draw multiple arrows from the **same parent class** to its subclasses.

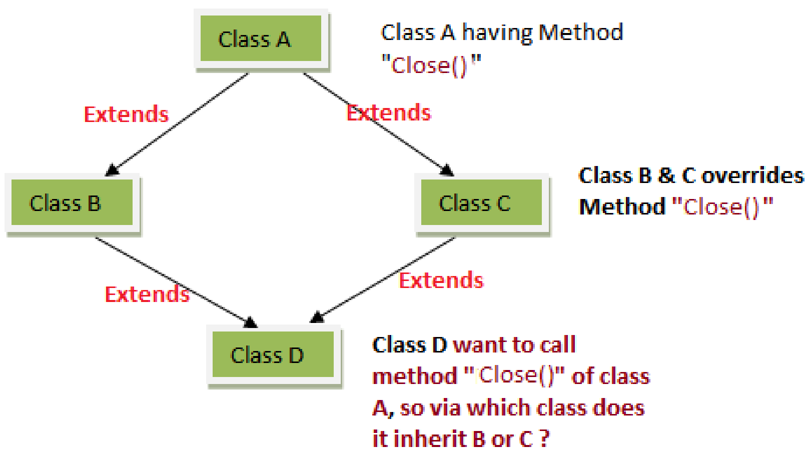
4. Hybrid Inheritance :

- **Definition:** Hybrid inheritance is a **combination of two or more types of inheritance** (e.g., single + multiple, or multilevel + hierarchical).
- It is called "hybrid" because it mixes different inheritance patterns.

5. Multiple Inheritance (Not in Java with classes) :

- A single subclass having arrows pointing to more than one parent class.
- In Java, show this only as a **conceptual diagram** (with a note: *not supported with classes*).
- Java allows it only via **interfaces**.

Deadly Diamond Problem



3.Does Java provide a **direct solution** to the **Deadly Diamond Problem**?

Ans : No, Java **does not support multiple inheritance with classes**.

- This means: **the diamond problem cannot occur in Java's class-based inheritance model**.
    - Instead of multiple inheritance, java provides alternative solution [Interfaces].

- Multiple inheritance is possible in Java through interfaces.

<u>**Today's Work**</u> : Design an **inheritance hierarchy** for calculating mileage of **Scooter, Bike, and Electric Scooter** in Java.